Phishing Email Detection & Awareness System

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1. Analyzed Phishing Evidence

This report presents a comprehensive breakdown of three intercepted phishing messages. These cases highlight multiple attack strategies designed to compromise both corporate systems and personal accounts. The specific samples cover a broad spectrum of threats, featuring:

Analyzed Fingings

The following sections break down each intercepted sample based on header intelligence, domain inspection, psychological indicators, and overall risk classification

Evidence 1: The Email System Administrator (Urgent Verification)

2. Email Header Analysis

From: System Administrator <steve@awoc.ca (mailto:steve@awoc.ca)>

To: Email User <emailuser@email.com>

Subject: Your Mail Box: Verify My Account

Date: Tue, 10 Mar 2026 17:44:33 +0100

Email Body Transcript

Dear email User,

There is a security alert on your EMAIL account.
Kindly click "Verify My Account" below to resolve this problem.

Verify My Account
http://indokegstes.tripod.com/

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http://indokegstes.tripod.com/

3. Identified Phishing Indicators

- Legitimate companies never use generic titles like "System Administrator" without branding.
- The domain awoc.ca is unrelated to major email providers (Gmail, Outlook, corporate domains)
- Creates panic to bypass rational thinking (classic social engineering). No specific account details.
- Uses free hosting (Tripod/Lycos) – legitimate banks/tech companies never use it.
- HTTP (not HTTPS) - no encryption, easy to intercept.
- Low-effort template – real security alerts are professionally written and branded.

4. Risk Classification

Risk Level: (HIGH)

Classification Details: Credential Harvesting / Enterprise Access. The attacker aims to steal user email account credentials, which could lead to loosing access to the email account.

Evidence 2: The Bank Update (Urgent Update)

1. **Email Header Analysis**

From: "USAA" usaa-offers@banking2.org

To: Bank User <victim@gmail.com>

Subject: IMPORTANT NEW MESSAGE FROM USAA

Date: Tue, 10 Mar 2026 17:44:33 +0100

Email Body Transcript

Dear Member,

We're making some exciting changes that will make your online banking experience very secure. We therefore request you to update your account. It is now important for you to update your full profile for your account safety. We need you to update your account information immediately.

click here to update your account now

The security of your personal information is always a priority at USAA. We work hard to help ensure your account information stays secure.

To learn how USAA protects your information, visit our Security Center on usaa.com. We value your business and are committed to keeping your accounts and personal information safe and secure.

Thank you,

USAA

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Banking2.org

2. **IDENTIFIED PHISHING INDICATORS**

- banking2.org" is unrelated and suspicious – real banks don't use random third-party domains.
- Creates false urgency to prevent thinking/checking, Real USAA updates are never forced via unsolicited email links.

- Links in phishing emails lead to counterfeit pages mimicking USAA login to steal username/password
- No visible URL here, but pattern matches known credential-harvesters.
- Low-quality template – professional banks use clean, branded HTML.

3. Risk Classification

Risk Level: (HIGH)

Classification Detail: Phishing (High confidence – clear intent to deceive and steal credentials)

Evidence 3: The PayPal False Detection (Urgent Update)

1. **Email Header Analysis**

From: PayPal security < paypal@e.paypal.co.uk >

To: PayPal User <victim@gmail.com>

Subject: Your Account Has Been Detected

Date: Tue, 10 Mar 2026 17:44:33 +0100

Email Body Transcript

Dear Customer,

Log in to your PayPal account as soon as possible we have recently encountered a problem in your account. When you created your PayPal account to make sure you're the account holder

Click the Activation link and Follow the instructions.

! [Notice] If this email was sent to you in Junk or Spam folder

Don't consider it as spam, this folder was mistakenly marked as spam by our new security update.

Thanks.

ACTIVATE ACCOUNT

Get more out of your account now is the time to make your PayPal account even more useful:

Account Overview Get up-to-date information about your payments.

Add or withdraw money from your PayPal account. See a detailed transaction history.

Yours sincerely,

PayPal

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paypal.co.uk

2. IDENTIFIED PHISHING INDICATORS

- "paypal.co.uk" is a deliberate misspelling to fool quick glances – a common typo squatting trick.
- Creates panic without specifics. Real PayPal alerts are detailed (e.g., exact transaction) and never demand immediate link clicks.
- "Click the Activation link" Leads to counterfeit PayPal login page to steal username/password
- Attempts to bypass filters by instructing user to mark as not spam – a red flag tactic used by phishers
- Mimics legitimate content to build trust, but inconsistencies (e.g., no real logo, generic sign-off) give it away.

3. Risk Classification

Risk Level: (HIGH)

Classification Detail: Phishing (High confidence – clear intent to deceive and steal credentials), The attacker intend to steal credential which leads to financial loss and compromised business.

➢ Prevention & Awareness Guidelines Key rules for all staff – based on analyzed phishing samples:

1. Verify Sender Identity
   Always reveal the real email address (hover/click sender name). Only trust official domains:@paypal.com, @usaa.com, etc.
   Flag typo squatted domains (e.g., paypal.co.uk, banking2.org) and free webmail accounts claiming to be official.

2. Ignore Urgent Threats
   Emails demanding immediate action ("verify now", "account on hold", "update profile immediately") are almost always scams. Never click links. Confirm requests via a known phone number or official website login.

3. Check Links carefully
   hover over every link/button. Destination must match the real domain (e.g., paypal.com, usaa.com).Red flags: free hosts (tripod.com), gibberish names, HTTP (not HTTPS), shortened URLs.

4. Never Open Unexpected Attachments
   Treat .doc, .xls, .zip files from unknown sources as dangerous — they often carry malware or ransom ware.

5. Verify Banking/Payment Changes by Phone
   Any notice of changed bank details or payment instructions must be confirmed by calling a known contact at the vendor — never via email reply or link.

6. Report – Don't Just Delete
   Use the "Report Phishing" button or forward to security@yourcompany.com. Reporting helps protect everyone.

7. Enable MFA / 2FA everywhere

8. Turn on multi-factor authentication for email, banking, and payment accounts. It blocks access even if passwords are stolen.